

# Daniel Lu

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## Education

**Carleton University, B.Sc. in Computer Science, Honours Program (Co-op)**

Ottawa, Canada

*AI & Machine Learning Stream - Dean's Honours List - Major CGPA: 4.0/4.0*

*Sep 2023 – Present*

- *Carleton University Entrance Scholarship (\$16,000)*

## Technical Skills

**Languages:** C/C++, Python, Java, JavaScript, TypeScript, C#, SQL, HTML, PHP, CSS

**Libraries and Frameworks:** TensorFlow, PyTorch, Node, React, Next.js, TailwindCSS, Flask, React Native, Selenium

**Platforms, Technologies, and Knowledge:** Software testing, Shell scripting, Git, AWS, Docker, Linux, DSA

## Projects

**Walk in the Park** | [Devpost](#) | [GitHub](#)

*April 2024, University of Toronto*

- Achieved **Best AI in Education Hack** in GenAI Genesis 2024 (out of **250+** participants).
- Used **React Native**, **Gemeni**, and **Flask** to build a **gamified mobile application** that aims to foster empathy, understanding, and cultural awareness towards local communities.

**TRACY: Tennis Realtime Analysis Coaching Systems** | [Devpost](#) | [GitHub](#)

*Feb 2024, Queen's University*

- Achieved **3rd Best Hack** in QHacks 2024 (out of **200+** participants).
- Developed an accessible, real-time **React web app** for **tennis analysis and coaching**, implementing **computer vision algorithms** and a **CRNN** powered by **OpenCV** and **TensorFlow**.
- Effectively tracks **rapid ball and human movements in 3D space** from a singular viewpoint and delivers personalized insights to enhance the skills of tennis enthusiasts.

**Melodica** | [Devpost](#) | [GitHub](#)

*Oct 2023, McGill University*

- Achieved the **Best Hack for All Arts** in McGill AI Hacks (*MAIS Hacks*) 2023 (out of **150+** participants).
- Crafted a **modern React web tool for musicians** that leveraged **Tensorflow** and other **open source neural networks** to separate instrument stems and provide other valuable utilities.

**Automatic Speech Recognition Neural Network** | [Read](#) | [GitHub](#)

*Jun 2022 - Jan 2023*

- Published a machine learning research paper on **automatic speech recognition** that investigated the performances of different **speech feature extraction algorithms** in ASR systems.
- Trained a **deep bidirectional GRU network** using **TensorFlow** to accurately transcribe continuous speech data using varying audio feature extraction algorithms, such as Mel spectrograms, Mel-Frequency Cepstral Coefficients (MFCCs), and discrete wavelet transforms.

## Experience

**Software Subteam Lead, Bot Developer, and Mentor**

*Jun 2021 – April 2024, Part-time*

*FIRST Robotics Team 8729: Spark Youth Robotics Club*

*Kanata, Ontario*

- **Organized online workshops** tailored toward children to spark interest in **robotics, software development, and engineering**
- Managed the **autonomous and lighting software subteams** during the FRC build season.
- Introduced team members to **WPILib**, **PID systems**, and the **GitHub workflow**; responsible for preparing weekly lesson plans.
- Became the **first team from Ottawa** to qualify for the FIRST Championships in **4 years, as rookies**.
- Noticed inefficiencies in managing club tasks across multiple platforms; used **Node.js** paired with **PostgreSQL** to craft a **general-purpose Discord bot** tailored towards streamlining various club tasks.

## Awards & Achievements

- **International Baccalaureate Diploma:** Awarded to students who graduated from the International Baccalaureate Diploma Programme (*Higher Level in Math, Physics, and English*)
- **Senior Canadian Computing Competition Distinction Award (Feb 2023):** Awarded to students with a comprehensive knowledge of informatics who performed to an exceptional standard (*35/75 - top 20%*)